RENA

Reliable Evaluation for Neural Unlearning Audits

Audit ID	RENA-AUDIT-0001
Algorithm Audited	gradient_ascent
Generated	2026-08-04 06:58 UTC
Dataset	64-feature, 10-class classification
Forget Set Size	135 samples
Retain Set Size	1212 samples

Executive Summary

Consensus Verdict: Inconclusive

Consensus score: **0.79** / 1.00 | Inter-module agreement: **0.41** / 1.00

• Verification modules disagree (agreement=0.41, below the 0.5 threshold), so no confident verdict can be issued even though the average forget score is 0.79.
• Largest disagreement: 'Confidence Analysis' scored 0.33 while 'Bootstrap Statistical Witness' scored 0.99 — these tests disagree on whether forgetting occurred.

Model Utility

Split	Accuracy
Retain set	99.8%
Test set	93.1%
Forget set	97.0%
Original model (test set, pre-unlearning)	95.8%

Verification Module Results

Module	Forget Score	Reliability	Raw Statistic
Membership Inference Attack	0.94	1.00	0.5304

Loss Distribution KS-Test	0.89	1.00	0.1111
Confidence Analysis	0.33	1.00	0.0335
Bootstrap Statistical Witness	0.99	1.00	0.2947

Membership Inference Attack: forget-set size (135) meets the reliability threshold
Loss Distribution KS-Test: forget-set size (135) meets the reliability threshold
Confidence Analysis: forget-set size (135) meets the reliability threshold
Bootstrap Statistical Witness: forget-set size (135) meets the reliability threshold

Consensus Weighting

Module	Weight in Consensus Score
Membership Inference Attack	25.0%
Loss Distribution KS-Test	25.0%
Confidence Analysis	25.0%
Bootstrap Statistical Witness	25.0%

Recommendations

• Increase forget-set size if possible, or run additional verification rounds with different random seeds to check whether disagreement is due to sampling noise rather than a genuine leak.

Methodology & Limitations

This audit reports statistical evidence of forgetting across four independently computed verification modules, combined via a reliability-weighted consensus engine that forces an 'Inconclusive' verdict whenever the modules disagree beyond a calibrated threshold, rather than averaging disagreement into false confidence. This is not a formal or cryptographic guarantee of data deletion — it is evidence, not proof. Verification power decreases with small forget-set sizes and with datasets/models where baseline accuracy is very high, since there is less room for a detectable statistical signal. Reliability discounts applied to specific modules for this audit are listed above and should be read alongside the verdict.